Design of a 2.4-GHz Direct-Conversion CMOS Receiver Signal Path Under a 100-nH On-Chip Inductance Budget

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***Abstract*—This paper presents the design and simulation of the RF-to-baseband signal path of a 2.4-GHz direct-conversion CMOS receiver in the Cadence gpdk045 process with a 1.8-V supply. The signal path consists of an inductively degenerated cascode low-noise amplifier (LNA), a double-balanced Gilbert-cell mixer, and a differential IF gain stage. The design is constrained to a total on-chip inductance of 100 nH, a single external input inductor with $Q \geq 50$, and a single bond-wire degeneration inductor with $Q = 15$. The standalone LNA achieves $S_{11} = -18.0$ dB, $S_{21} = 21.6$ dB, and a noise figure (NF) of 1.63 dB at 2.4 GHz, and is unconditionally stable. The complete signal path reaches a conversion gain of 37.1 dB, a 50-Ω input match whose -10-dB bandwidth (2.32–2.54 GHz) covers the 2.4-GHz ISM band, and a 700-Ω output resistance. Low-IF noise figure and linearity (IIP3 $= -33$ dBm) fall short of their targets; we trace the noise to flicker noise in the active mixer and discuss the resulting gain–noise–linearity trade-off.**

***Index Terms*—Direct-conversion receiver, low-noise amplifier, Gilbert-cell mixer, CMOS RF front-end, 2.4-GHz ISM band.**

I. Introduction

Direct-conversion (zero-IF) receivers translate the RF signal straight to baseband, removing the image-reject filter and IF stages of a superheterodyne receiver and making the architecture attractive for integrated CMOS radios [2], [1]. The same choice brings its own difficulties: because the wanted signal lands near DC, the receiver is sensitive to DC offsets, LO self-mixing, and the flicker noise of the mixer and baseband stages [2], [5].

This work designs the signal path of such a receiver for the 2.4-GHz ISM band (2.40–2.48 GHz), covering the LNA, the down-conversion mixer, and the IF gain stage. The LO path is assumed to supply a single-ended swing of 0.8 V peak-to-peak. The design targets and hardware constraints are listed in Table I; the most restrictive is the 100-nH budget on total on-chip inductance, which limits how much gain can be obtained from inductive loads.

The rest of the paper is organized as follows. Section II explains the choice of topology for each stage. Section III describes the sizing and iteration of the LNA, mixer, and IF amplifier. Section IV reports simulation results for the complete signal path, and Section V discusses the trade-offs behind the specifications that were not met.

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TABLE I
Design Targets and Constraints

Parameter	Target
Receiver gain	≥ 35 dB
RF band	2.40–2.48 GHz
Noise figure	≤ 4 dB
IIP3	> -8 dBm
Input resistance	50 Ω
Output resistance	< 800 Ω
Supply voltage	1.8 V
Total on-chip inductance	≤ 100 nH
External input inductor	one, $Q \geq 50$
Degeneration (bond-wire) inductor	one, $Q = 15$
On-chip capacitors	4% bottom-plate parasitic

TABLE II
Topology Selection

Stage	Topology	Rationale
LNA	Inductively degenerated cascode	Simultaneous noise and power match to 50 Ω without noisy resistors [3], [6]; the cascode improves reverse isolation and reduces the Miller effect.
Mixer	Double-balanced Gilbert cell	Provides conversion gain toward the 35-dB target and good LO–RF/LO–IF isolation; the balanced structure suppresses DC offsets and LO self-mixing [4].
IF stage	Differential amplifier	Supplies the remaining 10–15 dB of gain, preserves the differential mixer output, and sets $R_{\text{out}} < 800$ Ω.

II. Receiver Architecture

Fig. 1 shows the complete signal path. Table II summarizes the topology chosen for each stage and the reason for the choice.

A large LNA gain is the central choice of the architecture. By the Friis formula, the noise of the mixer and baseband stages is divided by the LNA power gain, so a high-gain LNA

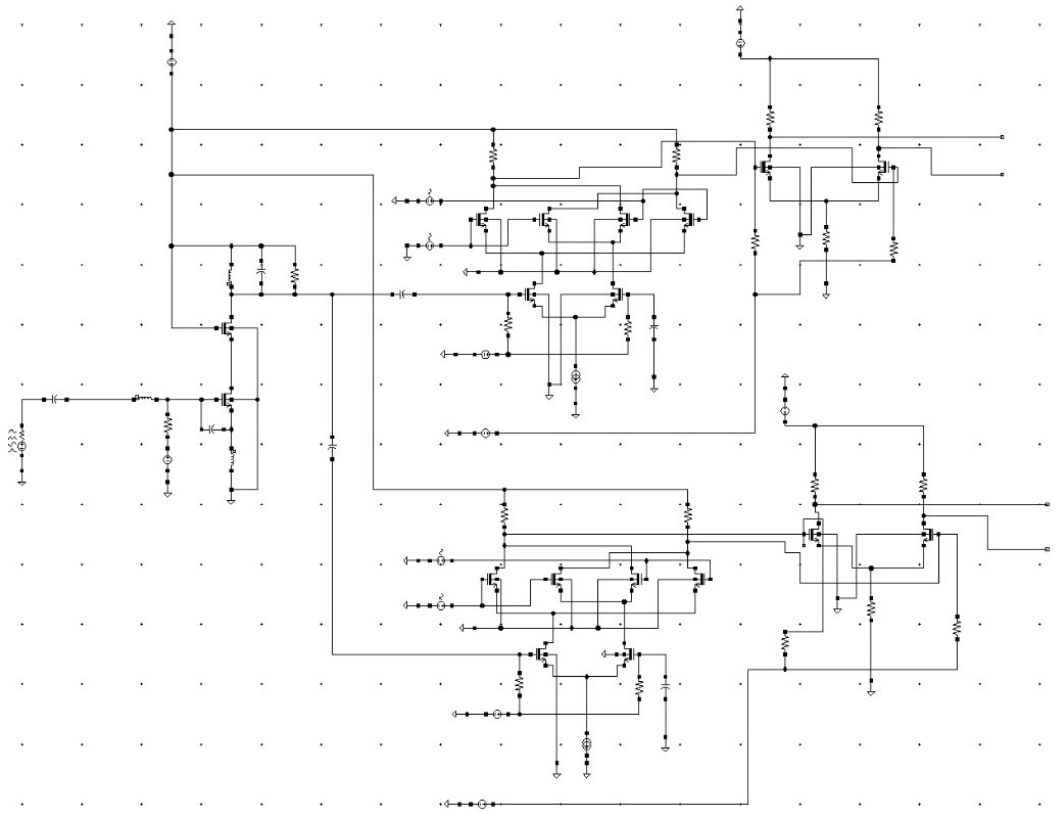


Fig. 1. Complete schematic of the 2.4-GHz direct-conversion receiver signal path: LNA, I/Q Gilbert-cell mixers, and differential IF gain stages.

is needed to keep the overall NF low while meeting the >35-dB gain target.

All transistors use the 2-V NMOS devices of gpdk045. Under a 1.8-V supply these devices provide more voltage headroom than the standard 1.1-V devices and avoid breakdown.

III. Circuit Design

A. Low-Noise Amplifier

Fig. 2(a) shows the LNA. The bias network was set so that both cascode transistors (M8, M9) operate in saturation, with transconductance g_m large enough for gain and drain current limited by the power budget.

With the initial component values, the S_{11} notch and the S_{21} peak fell outside the 2.4-GHz band. We swept the gate inductor L_G, the degeneration inductor L_0, the load inductor L_D, and the shunt capacitor at the lower transistor together, and used S_{11}, S_{21}, and the Smith chart jointly to locate the best matching point. Table III lists the evolution of the main values.

Increasing L_G from 10 nH to 29.5 nH adds enough inductive reactance to resonate with the gate–source capacitance C_{gs} at a lower frequency, moving the S_{11} notch to 2.4 GHz. Reducing L_0 from 1 nH to 0.15 nH compensates for the change in g_m/C_{gs} caused by the larger transistors, which keeps the real part of the input impedance near 50 Ω. Doubling L_D to 20 nH raises the AC impedance at the output node, and widening the transistors from 50 to 80 µm (10 to 16 fingers) increases g_m; together these push S_{21} above 15 dB. The unit finger width was held at 5 µm, so the larger device did not add excessive gate-resistance noise.

TABLE III
LNA Design Iteration

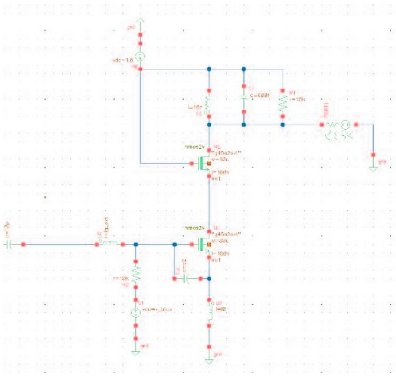
Parameter	Initial	Final
M8, M9 (W/L, fingers)	50 µm/180 nm, 10	80 µm/180 nm, 16
L_G (external)	10 nH	29.5 nH
L_0 (bond wire)	1 nH	0.15 nH
L_D (on-chip load)	10 nH	20 nH

Fig. 2(b) shows the final standalone LNA response. At 2.4 GHz, $S_{11} = -17.99$ dB, $S_{21} = 21.59$ dB, and NF $=$ 1.63 dB. The stability factor is $K_f = 1.08 > 1$ with $B_{1f} > 0$, so the LNA is unconditionally stable.

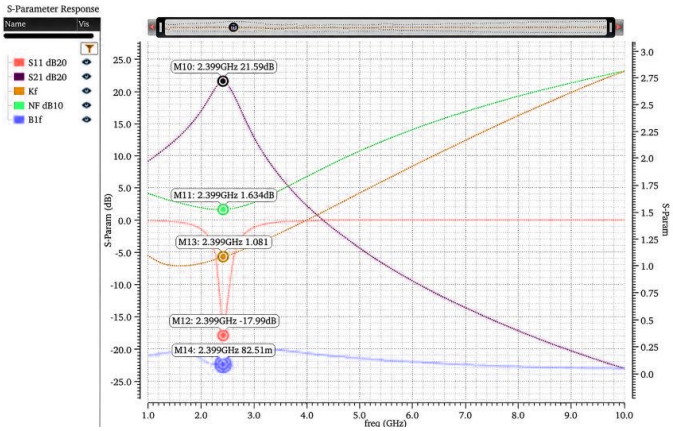
B. Gilbert-Cell Mixer

The mixer (Fig. 3) is a double-balanced Gilbert cell that converts the 2.4-GHz output of the LNA to baseband. Because the downconverted signal sits near DC, the sizing was driven mainly by flicker noise in the switching pair [5].

Switching pair (M1–M4). The devices were resized from 50 µm/180 nm to 10 µm/360 nm (10 fingers). The flicker-



(a)



(b)

Fig. 2. (a) Inductively degenerated cascode LNA. (b) Simulated S_{11}, S_{21}, NF, K_f, and B_{1f} of the final LNA; markers are at 2.399 GHz.

noise power spectral density scales inversely with gate area,

$$S_{1/f} = \frac{K}{C_{ox} W L f}, \tag{1}$$

so a longer channel lowers the $1/f$ noise that dominates the low-frequency SNR of a direct-conversion receiver.

Transconductance stage (M5–M6). The devices were resized from 60 µm/180 nm to 80 µm/360 nm (16 fingers). The larger width raises g_m to offset the loss in the switching stage, since the conversion gain of the mixer is approximately

$$G_{\text{conv}} \approx \frac{2}{\pi}\, g_{m5,6}\, R_L. \tag{2}$$

Load resistance. $R_L = 600\ \Omega$ gives sufficient conversion gain while leaving enough headroom under the 1.8-V supply to keep M1–M6 in saturation, which matters for linearity and for the output-resistance limit.

C. IF Amplifier

With the LNA and mixer in cascade, the simulated gain is about 27 dB at low IF (Fig. 4(a)), which is short of the 35-dB target. The LNA contributes less gain in cascade than standalone because of the loading of the mixer input. To close the gap, a differential amplifier was added after the mixer (Fig. 4(b)). This stage provides the remaining gain, keeps the signal differential as delivered by the Gilbert cell, and drives the output load. Table IV lists the final device sizes and component values of the signal path.

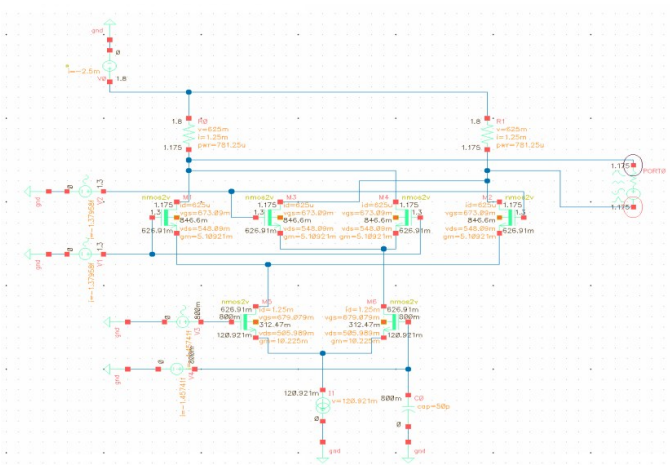


Fig. 3. Double-balanced Gilbert-cell mixer with DC operating points annotated.

TABLE IV
FINAL DEVICE SIZES AND COMPONENT VALUES

Stage	Devices	W/L	Fingers
LNA	M8, M9	80 µm/180 nm	16
Mixer	M1–M4	10 µm/360 nm	10
Mixer	M5, M6	80 µm/360 nm	16
IF amp	M0, M16	40 µm/360 nm	8

Component	Value	Component	Value
C_1	312 fF	R_L (mixer)	600 Ω
R	3 kΩ	R_{out}	700 Ω
C_5	10 pF	I_{SS}	350 µA
V_{amp}	0.6 mV	V_{DD}	1.8 V

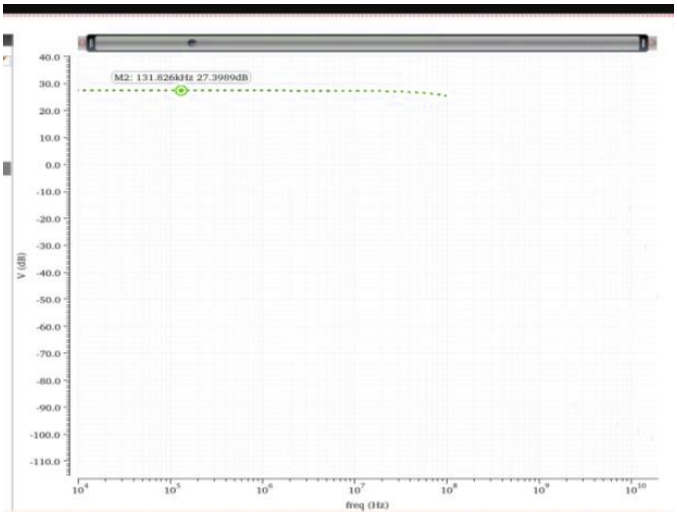
IV. Simulation Results

A. Gain and Noise Figure

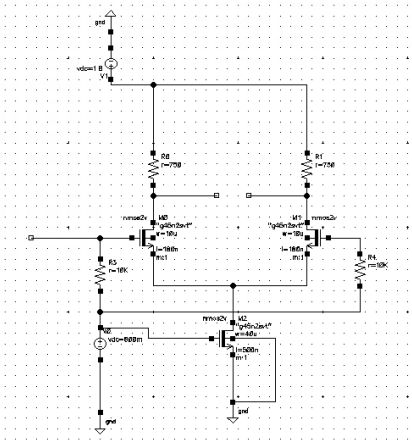
The first simulation of the complete signal path (Fig. 5) reached a conversion gain of 39.7 dB, above the 35-dB target. The NF, however, was 16.3 dB at a 10-kHz IF and fell only to 5.8 dB at 100 MHz. A noise-summary analysis attributed the low-frequency NF to flicker noise from the switching and load transistors of the active mixer.

B. Linearity

A two-tone test of the first full-chain design gave an IIP3 of -38 dBm, far below the -8-dBm target. We traded gain for linearity and noise by enlarging transistor areas and adjusting bias currents. This raised the IIP3 to -33 dBm and lowered the low-frequency NF floor, at the cost of conversion gain, which dropped to 34.8 dB. A final retuning brought the conversion gain of the complete path to 37.1 dB. In the final design the NF is about 15 dB at low IF and about 4 dB at high IF.



(a)



(b)

Fig. 4. (a) Simulated gain of the cascaded LNA and mixer versus IF frequency. (b) Differential IF amplifier.

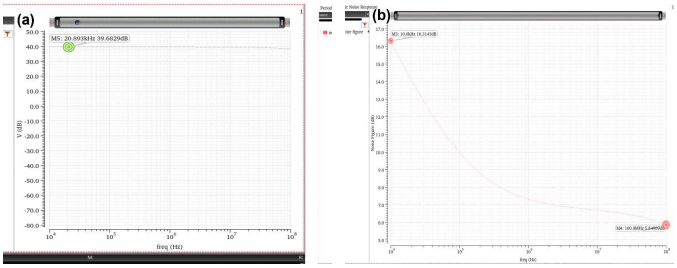


Fig. 5. First full-chain simulation, before the linearity and noise trade-off: (a) conversion gain versus IF frequency; (b) noise figure versus IF frequency.

C. Input Matching and Bandwidth

Fig. 6 shows the input return loss of the complete signal path. The minimum S_{11} is -19.06 dB at 2.426 GHz, meaning that less than 1.5% of the input power is reflected. The $S_{11} < -10$ dB bandwidth extends from about 2.32 to 2.54 GHz, roughly 220 MHz, which covers the whole ISM band; at the upper band edge of 2.48 GHz, S_{11} is still -12.91 dB.

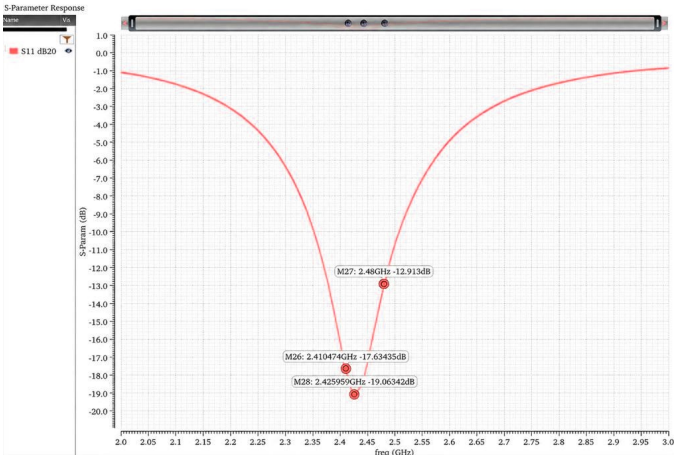


Fig. 6. Input return loss S_{11} of the complete receiver signal path.

TABLE V
DESIGN SUMMARY

Requirement	Target	Result	Status
Receiver gain	≥ 35 dB	37.1 dB	Pass
RF band	2.40–2.48 GHz	2.40–2.48 GHz	Pass
Noise figure	≤ 4 dB	$\approx$15 / $\approx$4 dB†	Partial
IIP3	> -8 dBm	-33 dBm	Fail
Input resistance	50 Ω	50 Ω	Pass
Output resistance	< 800 Ω	700 Ω	Pass
Supply voltage	1.8 V	1.8 V	Pass
On-chip inductance	≤ 100 nH	49.65 nH	Pass
External inductor	$Q \geq 50$	$Q \approx 50$	Pass
Bond-wire inductor	$Q = 15$	$Q = 15.08$	Pass
On-chip capacitors	4% parasitic	modeled	Pass

†Low IF / high IF.

D. Input and Output Resistance

The input resistance is matched to 50 Ω, and the output resistance is 700 Ω, within the 800-Ω limit.

E. Inductor Budget

The external gate inductor L_G (29.5 nH, 8.95-Ω series resistance) and the bond-wire degeneration inductor L_0 (150 pH, 150-mΩ series resistance) have quality factors of about 50 and 15.08 at 2.4 GHz, matching their respective constraints. The total on-chip inductance is 49.65 nH, below the 100-nH budget.

V. DISCUSSION

Table V compares the final design against the targets. Gain, input matching, bandwidth, output resistance, and all hardware constraints are met. Noise figure and IIP3 are not.

The two misses share a cause. The receiver gain, noise, and linearity pull against each other in this design. A high-gain LNA lowers the noise contribution of later stages, but it also delivers a larger signal to the mixer, whose switching pair and load then limit IIP3. Enlarging the mixer devices lowers their flicker noise but adds capacitance and reduces conversion gain, which the IF stage must recover. In this work we gave

priority to the mandatory gain threshold and to flicker-noise suppression, and accepted the shortfall in IIP3.

The low-IF noise figure points to the active mixer as the main limitation of the architecture. Approaches that avoid DC current in the switching pair, such as a passive current-commutating mixer followed by a transimpedance amplifier, are a natural next step for reducing flicker noise and improving linearity together [5], [1].

VI. Conclusion

We designed and simulated the signal path of a 2.4-GHz direct-conversion CMOS receiver in gpdk045 under a 1.8-V supply and a 100-nH on-chip inductance budget. The design reaches 37.1 dB of conversion gain, a 50-Ω input match over 2.32–2.54 GHz, and a 700-Ω output resistance while satisfying all inductor and capacitor constraints. Its noise figure and IIP3 fall short of the targets, and the analysis attributes both to the active Gilbert-cell mixer, which identifies the mixer topology as the first place to improve.

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